Input

$\mathcal{S}$

Select a query point $\mathbf{s}_i$

No

Traversed?

Yes

Output

$\mathcal{P}$

Generate detectors

Search contacts

Pair contacts

If $\mathbf{f}_i$ is face

1D detector (Cylinder)

If $\mathbf{f}_i$ is edge

2D detector (Cylindrical sector)

If $\mathbf{f}_i$ is corner

3D detector (Cone)

$\mathcal{D}_i$

l

d

θ_{sec}

θ_{cone}

(a.1)

(a.2)

(a.3)

$\mathcal{M}_i$

(b.1)

(b.2)

(b.3)

$\mathcal{C}_i$

(c.1)

(c.2)

(c.3)